

# Ratio & Proportion

Comprehensive Study Guide, Solved Examples (12) & 25 Practice MCQs with Answer Key

## 1. Comprehensive Theory & Concepts

### 1. What is a Ratio?

A ratio is a comparison of two quantities of the same kind by division. If  $a$  and  $b$  are two quantities of the same kind (with  $b \neq 0$ ), then the ratio of  $a$  to  $b$  is written as  $a : b$  or expressed as the fraction  $a/b$ .

#### Key Properties of Ratios:

- The quantities must be in the same units (e.g., compare meters to meters, not meters to centimeters directly without conversion).
- A ratio has no unit; it is a pure dimensionless number.
- Multiplying or dividing each term of a ratio by the same non-zero number does not alter the ratio (e.g.,  $2 : 3 = 4 : 6 = 20 : 30$ ).

### 2. Types of Ratios

- **Duplicate Ratio:** The ratio of squares of two numbers. Duplicate ratio of  $a : b$  is  $a^2 : b^2$ .
- **Sub-duplicate Ratio:** The ratio of square roots of two numbers. Sub-duplicate ratio of  $a : b$  is  $\sqrt{a} : \sqrt{b}$ .
- **Triplicate Ratio:** The ratio of cubes of two numbers. Triplicate ratio of  $a : b$  is  $a^3 : b^3$ .
- **Sub-triplicate Ratio:** The ratio of cube roots of two numbers. Sub-triplicate ratio of  $a : b$  is  $\sqrt[3]{a} : \sqrt[3]{b}$ .
- **Inverse / Reciprocal Ratio:** The inverse of  $a : b$  is  $b : a$ .
- **Compound Ratio:** The ratio obtained by multiplying two or more ratios together. Compound ratio of  $a : b$  and  $c : d$  is  $ac : bd$ .

### 3. What is a Proportion?

An equality of two ratios is called a proportion. If  $a : b = c : d$ , we write it as  $a : b :: c : d$ , which means  $a/b = c/d$ .

**Product of Extremes = Product of Means  $\Rightarrow a \times d = b \times c$**

- **Fourth Proportional:** If  $a : b = c : d$ , then  $d$  is the fourth proportional to  $a, b, c$ . ( $d = (b \times c) / a$ )
- **Third Proportional:** If  $a : b = b : c$ , then  $c$  is the third proportional to  $a$  and  $b$ . ( $c = b^2 / a$ )
- **Mean Proportional:** The mean proportional between  $a$  and  $b$  is  $\sqrt{ab}$ .

#### 4. Important Rules of Proportion

- **Componendo and Dividendo:** If  $a / b = c / d$ , then  $(a + b) / (a - b) = (c + d) / (c - d)$ .
- **Alternendo:** If  $a / b = c / d$ , then  $a / c = b / d$ .
- **Invertendo:** If  $a / b = c / d$ , then  $b / a = d / c$ .
- **Addendo:** If  $a / b = c / d = e / f = \dots$ , then each ratio equals  $(a + c + e + \dots) / (b + d + f + \dots)$ .

## 2. Solved Examples (12 Comprehensive Problems)

### Example 1: Basic Ratio Simplification

If  $A : B = 3 : 4$  and  $B : C = 8 : 9$ , find  $A : B : C$ .

**Solution:**

Make the term for  $B$  equal in both ratios.

$$A : B = 3 : 4 = (3 \times 2) : (4 \times 2) = 6 : 8.$$

Given  $B : C = 8 : 9$ .

Therefore, combining them directly gives  $A : B : C = 6 : 8 : 9$ .

### Example 2: Distributing an Amount

Divide ₹ 1,260 among A, B, and C in the ratio  $2 : 3 : 4$ . Find the share of B.

**Solution:**

Sum of ratio terms =  $2 + 3 + 4 = 9$ .

$$B's \text{ share} = (3 / 9) \times 1260 = (1 / 3) \times 1260 = ₹ 420.$$

Answer: B's share is ₹ 420.

### Example 3: Finding Added Quantity

Two numbers are in the ratio  $5 : 7$ . If 4 is added to each, the ratio becomes  $3 : 4$ . Find the numbers.

#### Solution:

Let numbers be  $5x$  and  $7x$ .

According to the question:  $(5x + 4) / (7x + 4) = 3 / 4$ .

$$4(5x + 4) = 3(7x + 4) \Rightarrow 20x + 16 = 21x + 12 \Rightarrow x = 4.$$

Numbers are  $5(4) = 20$  and  $7(4) = 28$ . (Numbers are 20 and 28).

### Example 4: Fourth Proportional

Find the fourth proportional to 6, 8, and 15.

#### Solution:

Let the fourth proportional be  $x$ .

Then  $6 : 8 :: 15 : x$ .

$$6 / 8 = 15 / x \Rightarrow 6x = 120 \Rightarrow x = 20.$$

Answer: Fourth proportional is 20.

### Example 5: Third Proportional

Find the third proportional to 16 and 36.

#### Solution:

Let the third proportional be  $x$ .

Then  $16 : 36 :: 36 : x$ .

$$16 / 36 = 36 / x \Rightarrow 16x = 36 \times 36 \Rightarrow 16x = 1296 \Rightarrow x = 81.$$

Answer: Third proportional is 81.

### Example 6: Mean Proportional

Find the mean proportional between 9 and 16.

#### Solution:

Mean proportional between  $a$  and  $b$  is  $\sqrt{ab}$ .

$$\text{Mean} = \sqrt{9 \times 16} = \sqrt{144} = 12.$$

Answer: Mean proportional is 12.

### Example 7: Income and Expenditure

The incomes of A and B are in the ratio **3 : 4** and their expenditures are in the ratio **2 : 3**. If each saves ₹ 2,000, find A's income.

**Solution:**

Let incomes be **3x** and **4x**, and expenditures be **2y** and **3y**.

$$3x - 2y = 2000 \text{ and } 4x - 3y = 2000.$$

Subtracting the two equations:  $(4x - 3y) - (3x - 2y) = 0 \Rightarrow x = y$ .

Substituting  $x = y$  in first equation:  $3x - 2x = 2000 \Rightarrow x = 2000$ .

A's income =  $3x = 3 \times 2000 = ₹ 6,000$ .

### Example 8: Coin Problem

A bag contains 50p, 25p, and 10p coins in the ratio **5 : 9 : 4**, amounting to ₹ 206. Find the number of 25p coins.

**Solution:**

Let number of coins be **5x**, **9x**, and **4x**.

$$\text{Total value in rupees} = (5x \times 0.50) + (9x \times 0.25) + (4x \times 0.10) = 2.50x + 2.25x + 0.40x = 5.15x.$$

$$5.15x = 206 \Rightarrow x = 206 / 5.15 = 40.$$

$$\text{Number of 25p coins} = 9x = 9 \times 40 = 360.$$

### Example 9: Mixture Ratio

In a mixture of 60 liters, the ratio of milk to water is **2 : 1**. What amount of water must be added to make the ratio **1 : 2**?

**Solution:**

$$\text{Milk} = 60 \times (2/3) = 40 \text{ liters; Water} = 60 \times (1/3) = 20 \text{ liters.}$$

Let **y** liters of water be added.

$$40 / (20 + y) = 1 / 2 \Rightarrow 80 = 20 + y \Rightarrow y = 60 \text{ liters.}$$

Answer: 60 liters of water must be added.

### Example 10: Equal Ratios Property

If  $a : b = c : d$ , prove that  $(a^2 + b^2) / (a^2 - b^2) = (ac + bd) / (ac - bd)$ .

#### Solution:

Let  $a/b = c/d = k$ , so  $a = bk$  and  $c = dk$ .

$$\text{LHS} = (b^2k^2 + b^2) / (b^2k^2 - b^2) = (k^2 + 1) / (k^2 - 1).$$

$$\text{RHS} = (bk \cdot dk + bd) / (bk \cdot dk - bd) = (bd k^2 + bd) / (bd k^2 - bd) = (k^2 + 1) / (k^2 - 1).$$

Hence, LHS = RHS.

### Example 11: Speed and Time Ratio

The speeds of three cars are in the ratio  $2 : 3 : 5$ . Find the ratio of time taken by them to cover the same distance.

#### Solution:

Since Time is inversely proportional to Speed ( $\text{Time} = \text{Distance} / \text{Speed}$ ), ratio of times =  $(1/2) : (1/3) : (1/5)$ .

LCM of 2, 3, and 5 is 30.

Multiply each by 30:  $(30/2) : (30/3) : (30/5) = 15 : 10 : 6$ .

Answer: Ratio of times is  $15 : 10 : 6$ .

### Example 12: Alloy Composition

An alloy contains copper and zinc in the ratio  $5 : 3$  and another alloy contains them in the ratio  $3 : 5$ . If equal quantities of both alloys are melted together, find the ratio of copper to zinc in the new alloy.

#### Solution:

In 1st alloy, Copper =  $5/8$ , Zinc =  $3/8$ .

In 2nd alloy, Copper =  $3/8$ , Zinc =  $5/8$ .

New Copper =  $5/8 + 3/8 = 8/8 = 1$ .

New Zinc =  $3/8 + 5/8 = 8/8 = 1$ .

Ratio of Copper to Zinc in new alloy =  $1 : 1$ .

## 3. Practice Questions (25 MCQs)

**Q1. If  $A : B = 2 : 3$  and  $B : C = 4 : 5$ , what is  $A : C$ ?**

(A)  $8 : 15$

(B)  $3 : 5$

(C)  $5 : 8$

(D)  $15 : 8$

**Q2. What is the fourth proportional to 4, 9, and 12?**

- (A) 24 (B) 27 (C) 36 (D) 18

**Q3. Find the third proportional to 9 and 15.**

- (A) 25 (B) 20 (C) 22.5 (D) 27

**Q4. What is the mean proportional between 6 and 24?**

- (A) 10 (B) 12 (C) 14 (D) 15

**Q5. Two numbers are in the ratio  $3 : 5$ . If 9 is subtracted from each, the new numbers are in the ratio  $12 : 23$ . The smaller number is:**

- (A) 27 (B) 33 (C) 45 (D) 55

**Q6. If  $x : y = 3 : 4$ , find the value of  $(7x + 3y) : (7x - 3y)$ .**

- (A)  $5 : 2$  (B)  $11 : 3$  (C)  $37 : 11$  (D)  $19 : 5$

**Q7. The sum of three numbers is 98. If the ratio of the first to the second is  $2 : 3$  and that of the second to the third is  $5 : 8$ , then the second number is:**

- (A) 20 (B) 30 (C) 48 (D) 38

**Q8. If ₹ 782 is divided into three parts proportional to  $1/2 : 2/3 : 3/4$ , the first part is:**

- (A) ₹ 182 (B) ₹ 204 (C) ₹ 224 (D) ₹ 250

**Q9. A mixture contains alcohol and water in the ratio  $4 : 3$ . If 5 liters of water is added to the mixture, the ratio becomes  $4 : 5$ . Find the quantity of alcohol in the given mixture.**

- (A) 10 liters (B) 12 liters (C) 15 liters (D) 20 liters

**Q10. Salaries of Ravi and Sumit are in the ratio  $2 : 3$ . If the salary of each is increased by ₹ 4,000, the new ratio becomes  $40 : 57$ . Sumit's current salary is:**

- (A) ₹ 38,000 (B) ₹ 42,000 (C) ₹ 45,000 (D) ₹ 51,000

**Q11. The duplicate ratio of  $3 : 4$  is:**

- (A)  $\sqrt{3} : 2$  (B)  $9 : 16$  (C)  $27 : 64$  (D)  $4 : 3$

**Q12. The sub-duplicate ratio of  $36 : 49$  is:**

- (A)  $6 : 7$  (B)  $7 : 6$  (C)  $1296 : 2401$  (D)  $18 : 24.5$

**Q13. If  $a : b = 5 : 7$ , then  $(3a + 5b) : (5a - 2b)$  equals:**

- (A)  $15 : 11$  (B)  $50 : 11$  (C)  $25 : 13$  (D)  $30 : 7$

**Q14. What number must be subtracted from each of the numbers 15, 17, 22, and 27 so that the remainders are in proportion?**

- (A) 1 (B) 2 (C) 3 (D) 4

**Q15. The ratio of the number of boys and girls in a school is  $8 : 5$ . If there are 160 girls, total strength of the school is:**

- (A) 256 (B) 416 (C) 384 (D) 400

**Q16. If  $x/5 = y/8$ , then what is  $(x + 5) : (y + 8)$ ?**

- (A)  $8 : 5$  (B)  $5 : 8$  (C)  $25 : 64$  (D)  $1 : 1$

**Q17. A sum of money is distributed among P, Q, and R in the ratio  $5 : 6 : 9$ . If P gets ₹ 500, what is the total sum?**

- (A) ₹ 1,500 (B) ₹ 1,800 (C) ₹ 2,000 (D) ₹ 2,200

**Q18. The compound ratio of  $2 : 3$ ,  $6 : 11$ , and  $11 : 2$  is:**

- (A)  $2 : 1$  (B)  $1 : 2$  (C)  $6 : 11$  (D)  $3 : 2$

**Q19. If  $3x + 5y : 3x - 5y = 5 : 1$ , then  $x : y$  is:**

- (A)  $2 : 1$  (B)  $3 : 2$  (C)  $5 : 3$  (D)  $4 : 1$

**Q20. Three numbers are in the ratio  $1 : 2 : 3$  and the sum of their cubes is 988. The numbers are:**

- (A) 2, 4, 6 (B) 3, 6, 9 (C) 4, 8, 12 (D) 1, 2, 3

**Q21.** A box contains ₹ 1 coins, 50p coins, and 25p coins in the ratio  $2 : 3 : 4$ , totaling ₹ 180. How many 50p coins are there?

- (A) 120 (B) 144 (C) 160 (D) 96

**Q22.** If  $a/b = c/d = e/f = 2/3$ , then  $(2a^2 + 3c^2 + 4e^2) / (2b^2 + 3d^2 + 4f^2)$  is equal to:

- (A)  $2/3$  (B)  $4/9$  (C)  $8/27$  (D)  $16/81$

**Q23.** In a zoo, the ratio of rabbits to pigeons is  $5 : 7$ . If their total heads are 144, how many pigeons are there?

- (A) 60 (B) 84 (C) 96 (D) 72

**Q24.** The ages of A and B are in the ratio  $5 : 7$ . Four years ago, their ages were in the ratio  $3 : 5$ . Find the present age of B.

- (A) 20 years (B) 28 years (C) 35 years (D) 42 years

**Q25.** If  $x = (\sqrt{5} + 1) / (\sqrt{5} - 1)$  and  $y = (\sqrt{5} - 1) / (\sqrt{5} + 1)$ , find the value of  $x^2 + xy + y^2 : x^2 - xy + y^2$ .

- (A)  $3 : 4$  (B)  $13 : 11$  (C)  $39 : 35$  (D)  $5 : 4$

### Answer Key (Practice Questions 1 to 25)

|         |          |          |          |          |
|---------|----------|----------|----------|----------|
| Q1: (A) | Q6: (C)  | Q11: (B) | Q16: (B) | Q21: (B) |
| Q2: (B) | Q7: (B)  | Q12: (A) | Q17: (C) | Q22: (B) |
| Q3: (C) | Q8: (B)  | Q13: (B) | Q18: (A) | Q23: (B) |
| Q4: (B) | Q9: (B)  | Q14: (C) | Q19: (C) | Q24: (B) |
| Q5: (B) | Q10: (D) | Q15: (B) | Q20: (A) | Q25: (C) |